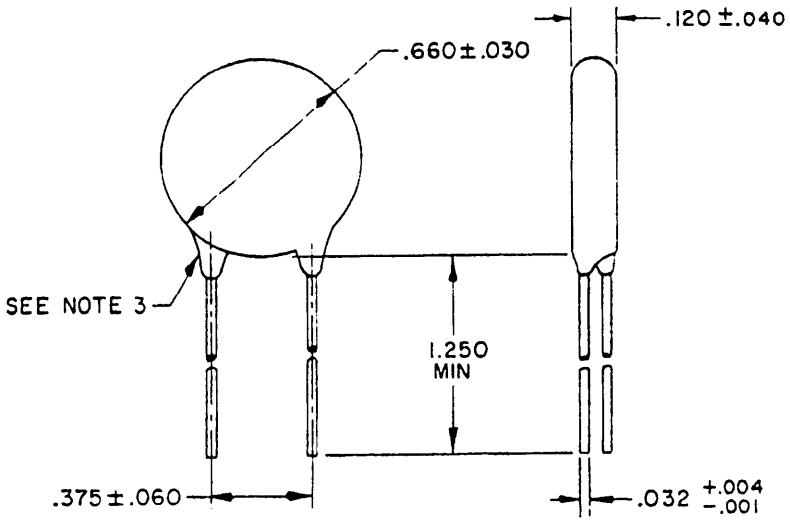
MILITARY SPECIFICATION SHEET

# CAPACITORS, FIXED, CERAMIC DIELECTRIC (GENERAL PURPOSE), STYLE CK63

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-11015.




| INCHES | MM |
|---|---|
| .001 | .03 |
| .004 | .10 |
| .030 | .76 |
| .032 | .81 |
| .040 | 1.02 |
| .060 | 1.52 |
| .120 | 3.05 |
| .375 | 9.53 |
| .660 | 16.76 |
| 1.250 | 31.75 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Insulating coating shall not extend more than .125 (3.18 mm) along lead wires measured from a tangent to the coating surface drawn perpendicular to the lead wires.

FIGURE 1. Dimensions and configuration.

(E) denotes changes.



FSC 5910

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TABLE I. Style CK63 characteristics.

| (E) Type designation 1/ | Rated voltage | Rated temperature and voltage-temperature limits | Capacitance | Capacitance tolerance |
|---|---|---|---|---|
| | Volts, dc | | pF | |
| CK63CZ681-- | 500 | CZ | 680 | K, M |
| CK63CZ821K- | 500 | CZ | 820 | K |
| CK63CZ102-- | 500 | CZ | 1,000 | K, M |
| CK63CZ152-- | 500 | CZ | 1,500 | K, M |
| CK63CZ222-- | 500 | CZ | 2,200 | K, M |
| CK63BX222-- | 500 | BX | 2,200 | K, M |
| CK63CZ332-- | 500 | CZ | 3,300 | K, M |
| CK63BX332-- | 500 | BX | 3,300 | K, M |
| CK63AW472M- | 1,000 | AW | 4,700 | M |
| CK63AY103M- | 1,000 | AY | 10,000 | M |
| CK63AW103M- | 500 | AW | 10,000 | M |

(E) 1/ Where applicable, the complete type designation will include an additional symbol to indicate capacitance tolerance. The type designation will also include the letter "E" to indicate an epoxy coated capacitor (when applicable) or "-" will be deleted for wax impregnated case.

REQUIREMENTS

Design and construction:

Dimensions and configuration - See figure 1.

(E) Case type - Disk, wax impregnated or epoxy coated (E).

Capacitance value - See table I.

Capacitance tolerance - ±10 percent (K) or ±20 percent (M) as shown in table I.

Rated temperature - -55° to +85°C (A); -55° to +125°C (B); or -55° to +150°C (C) as shown in table I.

Dielectric withstanding voltage:

Dielectric (DWV): Method 301 of MIL-STD-202.

Test voltage - 250 percent of rated voltage.

Body insulation: In accordance with MIL-C-11015.

Test potential - 1,300 volts dc.

Barometric pressure (reduced): Method 105 of MIL-STD-202, condition B (50,000 ft) for voltage-temperature limit Y, or 0.82 inch of mercury (80,000 ft) for voltage-temperature limits W, X, and Z.

Test potential - 150 percent of rated voltage.

Insulation resistance (IR): Method 302 of MIL-STD-202, condition B. 200,000 megohms, minimum.

Capacitance (Cap.): Method 305 of MIL-STD-202.

Test frequency - 1 kHz ±100 Hz.

Test voltage - 1.0 ±0.2 volt rms.



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Dissipation factor (DF): 1.5 percent, maximum (voltage-temperature limit X) or 2.0 percent, maximum (voltage-temperature limits W, Y, and Z).

Shock, medium impact: Method 205 of MIL-STD-202, condition C (50 G's).

Vibration, high frequency: Method 204 of MIL-STD-202, condition D (20 G's).

(E) Thermal shock and immersion:

Thermal shock: Method 107 of MIL-STD-202, condition A, except step 3 shall be at high test temperature.

Immersion: Method 104 of MIL-STD-202, condition B.

DWV - 250 percent of rated voltage.

IR - 150,000 megohms, minimum.

Salt spray (corrosion): Not applicable.

Terminal strength: In accordance with MIL-C-11015, 5 pounds applied.

Moisture resistance: Method 106 of MIL-STD-202.

DWV - 250 percent of rated voltage.

IR - 150,000 megohms, minimum.

Cap. - Within tolerance of table I value.

Solderability: Method 208 of MIL-STD-202. 2 terminals.

Resistance to soldering heat: Method 210 of MIL-STD-202, condition B.

IR - 200,000 megohms, minimum.

ΔCap. - ±5 percent of initial measurement.

ΔDF - 0.5 percent, maximum.

Voltage-temperature limits: In accordance with MIL-C-11015.

Life (at elevated ambient temperature): Method 108 of MIL-STD-202. 2,000 hours (for qualification); 250 hours (performance check); 1,750 hours (continuation test).

Test potential - 200 percent of rated voltage.

DWV - 250 percent of rated voltage (at 25°C).

IR - 100,000 megohms, minimum (at high temperature and 25°C).

Cap. - Within tolerance of table I value (at 25°C).

DF - 1.5 percent, maximum (voltage-temperature limit X) or 2.0 percent, maximum (voltage-temperature limits W, Y, and Z) (at 25°C).

Marking: In accordance with MIL-C-11015.



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Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - AS, OS, SH
Air Force - 11, 17
DLA - ES

User activities:
Navy - MC
Air Force - 19

Preparing activity:
Navy - EC

Agent:
DLA - ES

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INSTRUCTIONS. The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE
MIL C - 11015/12E

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

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DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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